author date

title

Teorema 1 (Criterio de Cauchy). *Sea* $\sum_n z_n \subset \mathbb{C}$ *una serie, entonces*

$$\sum_n z_n \textit{ converge} \iff \forall \varepsilon > 0 : \exists N_0 \in \mathbb{N} : \forall N \geqslant N_0, k \in \mathbb{N} : \left| \sum_{n=N}^{N+k} z_n \right| < \varepsilon.$$

***Demostración*:** $\sum_n z_n$ converge $\iff$ $(s_n)_{n\in\mathbb{N}}$ converge $\iff$ $(s_n)_{n\in\mathbb{N}}$ es de Cauchy

$$\iff \forall \varepsilon > 0 : \exists N_0 \in \mathbb{N} : \forall n, m = n + k \geqslant N_0 : |s_n - s_m| = \left| \sum_{k=n}^{n+k} z_k \right| < \varepsilon.$$

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